

Scala For DS

Practical no:-19

Roll No. : S019	Name: SUMEET KUMAVAT
Class : SY BSC IT	Date/Time of Submission : 18-08-2026

Aim:-Combine two plots (e.g., scatter and line plot) in a single visualization using Breeze-viz.

CODE:-

```
import breeze.linalg._
import breeze.plot._
object Practical19 extends App { new {
  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Sample X-axis values
  val yscat = DenseVector(2.0, 4.0, 1.0, 8.0, 7.0) // Y values for scatter
  val yline = DenseVector(1.5, 3.0, 4.5, 6.0, 7.5) // Y values for line plot
  val fig = Figure() // Create a new figure
  val plt = fig.subplot(0)
  plt += plot(x, yscat, '.', colorcode = "blue")
  plt += plot(x, yline, '-', colorcode = "red")
  plt.title = "Scatter + Line Plot"
  plt.xlabel = "X-axis"
  plt.ylabel = "Y-axis"
  fig.refresh()
}
```

```

import breeze.linalg._
import breeze.plot._
object Practical19 extends App { new {
  val x = DenseVector (1.0, 2.0, 3.0, 4.0, 5.0,6.0,7.0,8.0,9.0,10.0)
  val yscat = DenseVector (1.0, 2.0, 3.0, 4.0, 5.0,6.0,7.0,8.0,9.0,2.0)
  val yline = DenseVector(2.0, 2.0, 2.0, 2.0, 2.0,2.0,2.0,2.0,2.0,2.0)
  val fig = Figure()
  val plt = fig.subplot(0)
  plt += plot(x, yscat, '.', colorcode = "blue")
  plt += plot(x, yline, '-', colorcode = "red")
  plt.title = "Scatter + Line Plot"
  plt.xlabel = "X-axis"
  plt.ylabel = "Y-axis"
  fig.refresh()
}

```

OUTPUT:-

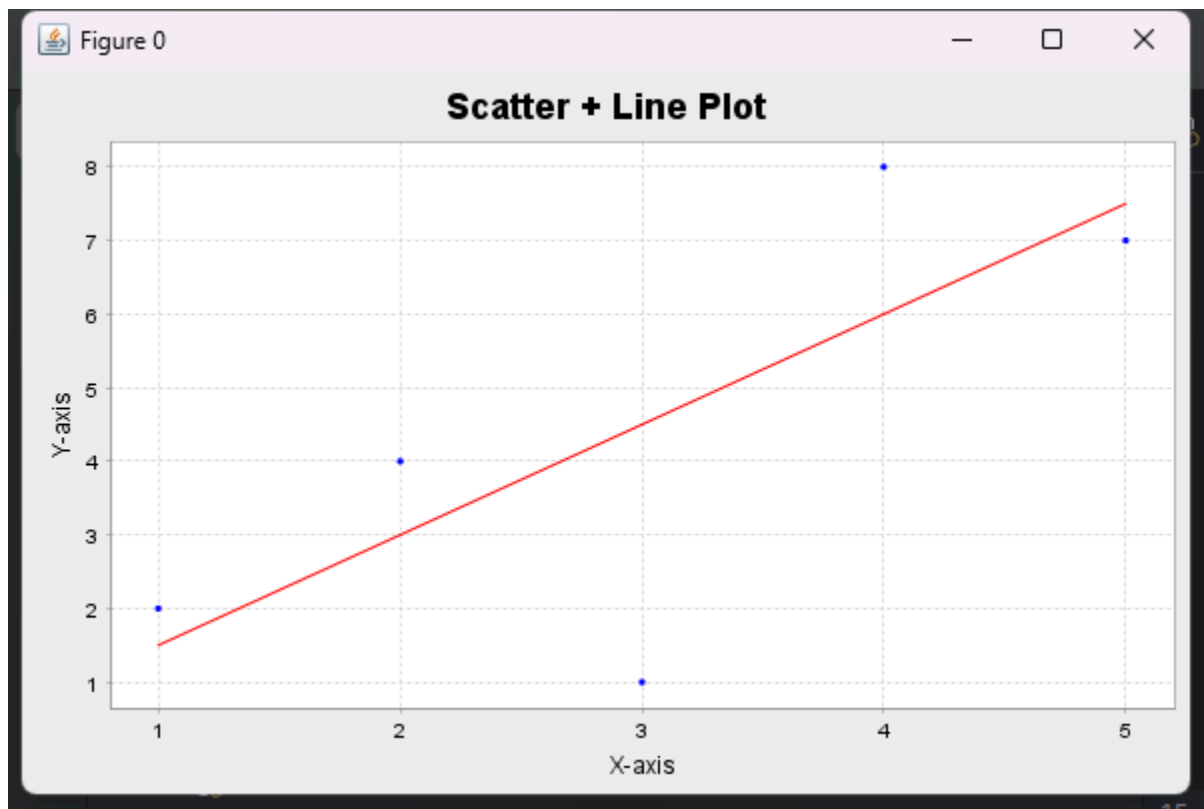
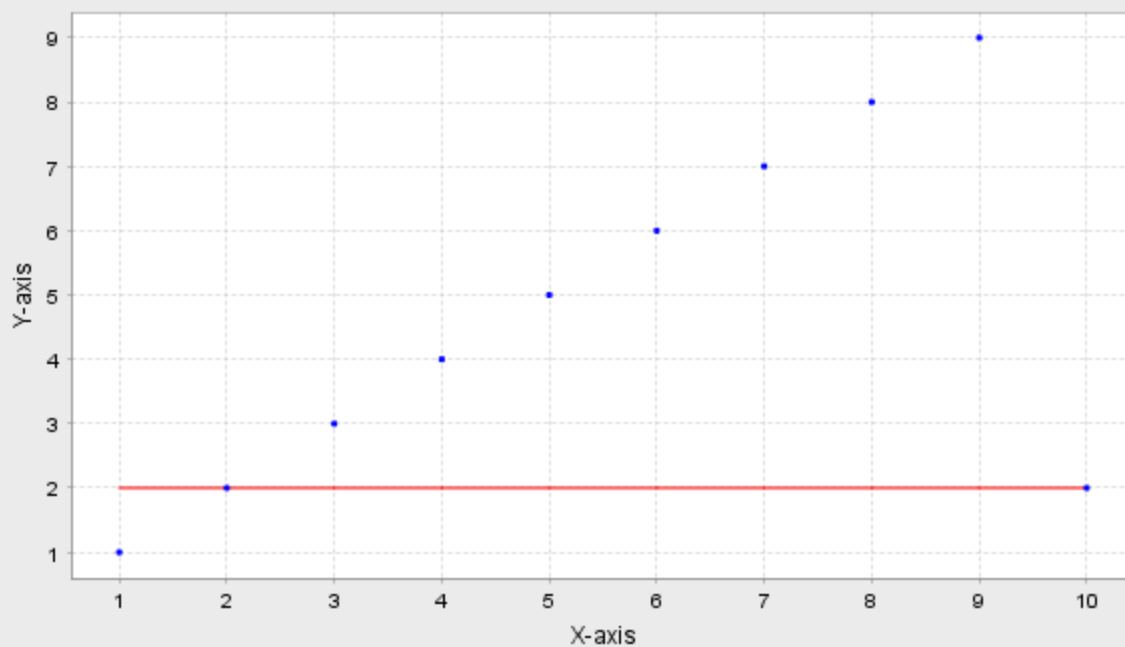


Figure 0

Scatter + Line Plot



Scala For DS

Practical no:-20

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Aim:- Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.

CODE:-

```
import scala.io.Source

object Practical extends App { new *
  val file = Source.fromResource("iris.csv")
  val data = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList
  file.close()

  val count = data.length
  val mean = data.sum / count
  val min = data.min
  val max = data.max
  val variance = data.map(x => math.pow(x - mean, 2)).sum / count
  val std = math.sqrt(variance)

  println(s"Count: $count")
  println(f"Mean: $mean%.2f")
  println(f"Minimum: $min%.2f")
  println(f"Maximum: $max%.2f")
  println(f"Standard Deviation: $std%.2f")
  println("\nSumeet S019")
}
```

OUTPUT;-

```
"C:\Program Files\Java\jdk1.8.0_181\bin
Count: 150
Mean: 75.50
Minimum: 1.00
Maximum: 150.00
Standard Deviation: 43.30

Sumeet S019

Process finished with exit code 0
|
```

Scala For DS

Practical no:-21

Roll No. : S019	Name: SUMEET KUMAVAT
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Aim:-Join two CSV files in Spark DataFrames based on a common column and write the output to a file.

Code:-

```
build.sbt  file1.csv  file2.csv  Practical21.scala ×
1  import org.apache.spark.sql.SparkSession
2
3  object Practical21 extends App { new *
4      val s = SparkSession.builder()
5          .appName("JoinTwoCSV")
6          .master("local[*]")
7          .getOrCreate()
8
9      val d1 = s.read
10         .option("header", "true")
11         .csv("File1.csv")
12
13     val d2 = s.read
14         .option("header", "true")
15         .csv("File2.csv")
16
17     val j = d1.join(d2, Seq("id"), "inner")
18     j.show()
19     j.write.option("header", "true")
20     s.stop()
21 }
22
```

Output:-

```
+---+-----+-----+
| id|  name|score|
+---+-----+-----+
| 1| Sumeet| 90|
| 2|Harshal| 70|
| 3| Dhruv| 85|
| 4| Niraj| 80|
+---+-----+-----+
```

```
Sumeet S019
```

```
Process finished with exit code 0
```